

TAO DENG

Ph.D. Student in Data Science

Shenzhen, Guangdong, China

☎ (+86) 182-7300-1407 | ✉ taodeng@link.cuhk.edu.cn | 🏠 <https://torydeng.github.io/> | 📺 ToryDeng | 🆔 0000-0001-7401-311X | 🎓 Tao Deng

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)

Ph.D. Student in Data Science

Advisors: [Prof. Hao Wu](#) and [Prof. Tianwei Yu](#)

Shenzhen, Guangdong, China

Sep. 2022 - Expected Jun. 2027

Zhongnan University of Economics and Law (ZUEL)

B.Sc. in Data Science and Big Data Technology

Worked under the supervision of [Prof. Xiaobo Sun](#)

Wuhan, Hubei, China

Sep. 2018 - Jun. 2022

Research Interests

- Developing scalable statistical and artificial intelligence (AI) methods for analyzing single-cell and spatial omics.
- Integrating multi-omics data to identify biologically relevant genes and infer gene-gene interactions.

Publications

† indicates equal contribution.

Working Papers

- [1] [Tao Deng](#), Zhe Yu, Xiaobo Sun, Tianwei Yu, and Hao Wu (2026). *PopSVG enables scalable detection of spatially variable genes in population-level spatial transcriptomics*. **Under second-round revision at Communications Biology** [[Code](#)]
- [2] Rulan Wang[†], Yingyuan Xie[†], [Tao Deng](#), Zhang Zhang, Guan-Zheng Luo, and Hao Wu (2026). *MultiNano: transformer-based multi-label prediction and interpretation of seven RNA modifications from nanopore direct sequencing*. **Under first-round revision at Briefings in Bioinformatics**

Peer-Reviewed Papers

- [1] [Tao Deng](#)[†], Mengqian Huang, Kaichen Xu, Yan Lu, Yucheng Xu, Siyu Chen, Nina Xie, Qiuyue Tao, Hao Wu, and Xiaobo Sun[†] (2025). *LEGEND: Identifying Co-expressed Genes in Multimodal Transcriptomic Sequencing Data*. **Genomics, Proteomics & Bioinformatics**, 23(4), qzaf056. [[DOI](#)] [[Code](#)]
- [2] Luxiao Chen[†], Zhenxing Guo[†], [Tao Deng](#), and Hao Wu (2024). *scCTS: identifying the cell type-specific marker genes from population-level single-cell RNA-seq*. **Genome Biology**, 25(1), 269. [[DOI](#)] [[Code](#)]
- [3] [Tao Deng](#)[†], Siyu Chen, Ying Zhang, Yuanbin Xu, Da Feng, Hao Wu, and Xiaobo Sun[†] (2023). *A cofunctional grouping-based approach for non-redundant feature gene selection in unannotated single-cell RNA-seq analysis*. **Briefings in Bioinformatics**, 24(2), bbad042. [[DOI](#)] [[Code](#)]

Honors & Awards

2022 **Outstanding Bachelor's Thesis Award**, Zhongnan University of Economics and Law
2021 **National Encouragement Scholarship**
2019 **National Encouragement Scholarship**

Wuhan, China
Wuhan, China
Wuhan, China

Academic Service

2024 **Invited Reviewer**, Genomics, Proteomics & Bioinformatics
2022 – **Lab Server Maintenance**
Present

Teaching Experience

STA-4001 Stochastic Processes
Leading Teaching Assistant

CUHK-Shenzhen
Fall 2025

MAT-2041 Linear Algebra and Applications

Leading Teaching Assistant

CUHK-Shenzhen

Spring 2025

STA-2001 Probability and Statistics I

Teaching Assistant

CUHK-Shenzhen

Fall 2024, Spring 2023

STA-2001H Honours Probability and Statistics I

Teaching Assistant

CUHK-Shenzhen

Fall 2022